

CS482 Assignment 1 - Readme

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=== How to Run ===

```
python assn1-part1.py in.fasta out.txt
```

=== Algorithm Overview ===

Fit alignment is like global alignment, but T's prefix and suffix can be skipped for free. So S must fully align, but we only use a substring of T.

=== What Each Part Does ===

1. readFasta(filename)

- reads the simplified FASTA file
- extracts the 2nd and 4th lines as sequences S and T

2. buildDPTable(S, T)

- creates a $(m+1) \times (n+1)$ DP table
- first column: $D[i][0] = i * \text{INDEL}$ (S prefix must pay gap cost)
- first row: $D[0][j] = 0$ (T prefix is free to skip, this is the key difference from global alignment)
- fills table using: $D[i][j] = \max$ of three choices (diagonal, up, left), each cell takes $O(1)$ time

3. findAnswer(D, m, n)

- scans the last row $D[m][0..n]$ to find maximum score

- this is because T suffix is also free to skip, so best answer can end at any column j
- records the column index for backtracking

4. backtrack(D, S, T, endJ)

- starts from (m, endJ) instead of (m, n)
- traces back through the DP table to reconstruct alignment
- stops when i=0 (S is fully consumed)

5. writeOutput(filename, score, alignS, alignT)

- writes score and alignment strings to output file

=== Time Complexity ===

$O(mn)$ where $m = \text{len}(S)$, $n = \text{len}(T)$

- buildDPTable: two nested loops over m and n, each cell $O(1) \rightarrow O(mn)$
- findAnswer: one scan through last row $\rightarrow O(n)$
- backtrack: at most $m+n$ steps $\rightarrow O(m+n)$
- total: $O(mn) + O(n) + O(m+n) = O(mn)$

=== Space Complexity ===

$O(mn)$

- DP table D: $(m+1) \times (n+1) \rightarrow O(mn)$
- output alignment strings: at most $m+n$ characters $\rightarrow O(m+n)$
- total: $O(mn) + O(m+n) = O(mn)$